

SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1714

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5

Time Duration: 1 Hour
Total Marks:30

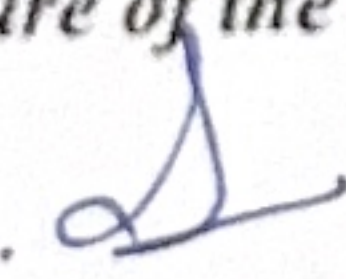
Annexure A

Resolution Algorithm Implementation

Code of Conduct:

I DILIP.S.V. (192424426) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

S.V. 





Job interview outcome
Knowledge - Base.

1) If an applicant passes the interview, then
the applicant is hired.

2) Farhan passed the interview.

a) Propositional Logic.

Let

* P = Farhan passes the interview

* H = Farhan is hired

Rule:

$$P \rightarrow H$$

Fact.

P

b) CNF conversion

using $P \rightarrow H = \neg P \vee H$:

$$\neg P \vee H$$

Fact.

c) We gave the goal

Goal

H

Negated goal:

$\neg H$

d) Resolution.

Resolve.

$\neg PH$

with

$\cdot P$

This given

H

Now Resolve.

H

with:

$\neg H$

Result.

Conclusion.

Since the empty clause is obtained

Farhan is hidden.

(

Attendance. ~~on~~ Credit.

Knowledge Base.

1) If a Student. attends all classes, then he.
Student get attendance credit.

2) meena attended all classes.

a) Propositional Logic.

Let

* A = meena attended all classes.

* C = meena get attendance credit

$$A \rightarrow C$$

A

b) CNF

$$\neg A \vee C$$

fact

A

c) negated goal.

goal.

C

Negated goal.

d) Resolution.

from

$\neg A \vee C$

and:

A

we derive

C

then

$C \rightarrow C$

given.



conclusion.

Therefore, maeha get attendance. credit

Car Rental Eligibility.

Let:

* L = Driver has valid license.

* R = Driver

Knowledge Base.

$L \rightarrow R$

L (Bures)

CHF

$\neg L \vee R$.

$\neg (Bures)$

gated Coal.

R (Buresh)

solution.

$L \vee R, L \Rightarrow R.$

$\neg R \Rightarrow$

Conclusion: Suraj can rent a car.

Post. Achievement. Award.

et.

- S = Player scores about 90%
- G = Player wins gold medal.

Knowledge Base.

$S \rightarrow G.$

S (Divya)

ENR

$\neg S \vee G$

S (Divya)

Not gated Goal.

$\neg G$ (Divya)

Conclusion: Divya win a gold medal.

5) Order shipping. Confirmation.

Let:

* $P = \text{Payment verified}$

* $C = \text{Order confirmed.}$

* $S = \text{item shipped.}$

$P \rightarrow C$

$C \rightarrow S$

P

CNF

$\neg P \vee C$

$\neg C \vee S$

P

negated goal

$\Rightarrow \neg P \vee C$

and:

P

we got

C

Then

$\neg C \vee S.$

Finally.

$S, \neg S = \square$

conclusion:

the item is shipped.